MOLAVE
TRADING INC.

ISO 9001:2015

January 5, 2022

**PARTIDO STATE UNIVERSITY**
Camarines Sur

**Attention:** **ENGR. KIER P. GASGA**
Project Leader

Dear Engr. Gasga,

As the authorized distributor of Agilent Technologies and in response to your request, we are pleased to offer the following:

| Item No. | Quantity | Part Number | Description |
|---|---|---|---|
| 1 | 1 unit | G6860A | **Agilent Cary 60 UV-VIS Spectrophotometer**<br><br>The Agilent Cary 60 UV-Vis spectrophotometer is efficient, accurate and flexible, and is designed to meet both current and future measurement needs. The proven, robust design of the Cary 60 comprises a double beam, Czerny-Turner monochromator, 190–1100 nm wavelength range, 1.5 nm fixed spectral bandwidth, full spectrum Xenon pulse lamp single source with exceptionally long life, dual silicon diode detectors, quartz overcoated optics, scan rates up to 24,000 nm/min, 80 data points/sec maximum measurement rate, non-measurement phase stepping wavelength drive, room light immunity, central control by PC with Microsoft® Windows® operating system. Supported by GLP software, optional 21 CFR Part 11 capable software, and dedicated instrument validation software which includes pharmacopeia test suites.<br><br>  <br><br>**Features:**<br>• Xenon pulse lamp source<br>- Eliminates photobleaching while allowing the highest quality data to be collected over the complete UV-Vis |

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| | | | range — all at the lowest cost of ownership due to an exceptionally long lamp life<br>• Room light immunity<br>• 1.5 nm fixed spectral bandwidth<br>• 190–1100 nm wavelength range<br>• 24,000 nm/min maximum scan rate<br>• 80 data points/second maximum measurement rate<br>• Photometric range up to 4 Abs<br>• Focused beam measuring 1.5 x 1.0 mm<br><br>**Technical Specifications:**<br>• **Source:** Unique full-spectrum Xenon flash lamp (80 Hz) with typical lifetime of 10 years (guaranteed 3 years)<br>• **Monochromator:** Czerny-Turner<br>• **Grating:** Holographic, 27.5 x 35 mm, 1200 lines/mm, blaze angle 8.6° at 240 nm<br>• **Beam splitting system:** Beam splitter<br>• **Detectors:** 2 silicon diode detectors for simultaneous sample beam and reference beam measurements<br>• **Optical design:** Double beam Czerny-Turner monochromator<br>• **UV-Vis limiting resolution (nm):** ≤ 1.5 nm<br>• **Toluene/hexane limiting resolution (EP/BP and TGA test):** ≥ 1.92<br>• **Stray light (%T)**<br>At 198 nm (12 g/L KCl, TGA & BP/EP method) ≤ 0.187%<br>At 220 nm (10 g/L NaI ASTM method) ≤ 0.018%<br>At 370 nm (50 mg/L $NaNO_2$) ≤ 0.008%<br>• **Wavelength range (nm):** 190–1100 nm<br>• **Wavelength accuracy (nm):** ± 0.06 at 541.94 nm<br>• **Wavelength reproducibility (nm):** ± 0.01 nm<br>• **Photometric accuracy (Abs)**<br>Using NIST 930E fi lters at 1 Abs ± 0.0007 Abs<br>• **Photometric range (Abs):** ± 4.0 Abs<br>• **Photometric display:** ± 9.9999 Abs, ± 200.00 %T<br>• **Photometric reproducibility (Abs)**<br>- Using NIST 930E fi lters, at 465 nm, 2 s SAT<br>Maximum deviation at 1 Abs < 0.004 Abs<br>Standard deviation for 10 measurements < 0.00050 Abs<br>- Using NIST 930E fi lters, at 546.1 nm, 2 s SAT<br>Maximum deviation at 0.5 Abs < 0.003 Abs<br>Standard deviation for 10 measurements < 0.0030 Abs<br>• **Photometric stability (Abs/hour)**<br>500 nm, 10 s SAT < 0.0004 Abs |

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| | | | • **Photometric noise (Abs, RMS)**<br>500 nm, 1 s SAT<br>- At 0 Abs < 0.00002 Abs<br>- At 1 Abs < 0.00012 Abs<br>- At 2 Abs < 0.0011 Abs<br>260 nm, 1 s SAT<br>- At 0 Abs < 0.00002Abs<br>• **Baseline flatness (Abs):** 200 to 850 nm, baseline corrected 0.0004 Abs<br>• **Compartment size (width x depth x height):** 130 mm x 523 mm x 123 mm<br>- Note that sample compartment can be left open during measurement due to room light immunity of Cary 60<br>• **Sample compartment access:** Top and front<br>• **Spectral bandwidth (nm):** Fixed at 1.5 nm (approximately)<br>• **Signal averaging (seconds):** 0.0125–999 s<br>• **Maximum scan rate (nm/min):** 24,000 nm/min<br>• **Slew rate (nm/min):** 24,000 nm/min<br>• **Data interval (nm):** 0.15–5.0 nm<br>• **Repetitive scanning:** 4800 data points per minute, maximum number of cycles: 999, maximum cycle time (min): 9999 |
| 2 | 1 unit | G6861AA | **Agilent Cary WinUV Software**<br><br>Compatible with:<br>• Microsoft Windows 10 Pro or Enterprise 64-bit<br>• Japanese, Simplified Chinese, Spanish and Portuguese<br>Provides:<br>• Scan, Reads, Kinetics (single or multi-wavelength and scanning) applications<br>• Validate application for performance testing |
| 3 | 1 ea | G6861AA#325 | Cary WinUV Academic Option |
| 4 | 1 pair | 6610000800 | 3.5mL, 10mm Pathlength Quartz Cuvette (Pair) |
| 5 | 1 unit | LOCAL | Desktop Computer, Windows 10, 64-bit |
| 6 | 1 unit | LOCAL | Printer |
| 7 | 1 unit | LOCAL | AVR 2KVA |
| **TOTAL PRICE (Item 1-7) .....Php 687,872.28** | | | |
| **OPTIONAL FIBER OPTICS ACCESSORIES** | | | |
| 8 | 1 unit | G6866A | **Dip Probe Coupler with doors** |

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| | | |  <br>Fiber optic probes must be ordered separately. |
| 9 | 1 unit | 7910036400 | **316 Stainless steel fixed (10 mm) pathlength probe**<br>• 600 um thick silica fibers<br>• SMA connector<br>• -50 to 200 deg temperature range<br>• 200 – 1100 nm wavelength range<br>• 78 cm long fiber cable<br>• 6.38 mm diameter |
| **TOTAL PRICE (Item 8-9) ... ..... Php 208,644.04** | | | |
| **TOTAL PACKAGE PRICE (Item 1-9) .....Php 896,516.32** | | | |

**TERMS OF PAYMENT** : Immediate

**DELIVERY** : Sixty-to-Ninety (60-90) Calendar days upon receipt of approved P.O.

**PRICE VALIDITY** : 120 Calendar days after receipt of offer, thereafter subject to final confirmation.

**TRAINING** : Offer includes training on-site upon installation by our Application Specialist.

**INSTALLATION** : To be done on-site by our factory trained Service Engineers and Application Specialist.

**SERVICE GUARANTEE** : As our commitment to service and technical excellence to our customers, we commit to respond within forty-eight (48) hours after confirmed receipt of call

**WARRANTY** : Offer includes one (1) year warranty under normal use of the instrument. Consumable parts are **NOT COVERED** by warranty.

**REMARKS** : Standard VAT INCLUDED.

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Thank you so much and we hope to receive your most valued order soon.

Yours very truly,

**MOLAVE TRADING INC.**

**AISAH D. PEÑARUBIA, RCh**
Sales Supervisor

ELD/EAA/MCB

**ACERO BUILDING** 891 E. Delos Santos Ave., Quezon City, Philippines 1103
Tel. Nos.: 3411-0360 to 64, 8924-2225 Fax: (632) 924-2094 Email: molave@acergroup.org

MOLAVE
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ISO 9001:2015

January 5, 2022

PARTIDO STATE UNIVERSITY
Camarines Sur

Attention: **ENGR. KIER P. GASGA**
Project Leader

Dear Engr, Gasga,

As the authorized Philippine distributor of Thermo Fisher Scientific, we are pleased to offer the following:-

One (1) unit **Thermo Scientific™ OGS60 Cat No. 51028139**
**Heratherm™ General Protocol Oven**



**Features:**

- Operating temperatures from 50° to 250°C
- Automatic over-temperature alarm system
- Low overall energy consumption
- Built in timer
- Corrosion-resistant stainless-steel chambers with rounded corners
- Large, easy to read vacuum fluorescent display
- Easy to use, microprocessor-controlled touch button operation
- Doors can be opened over 180°C
- Choice of gravity or mechanical convection models

**ACERO BUILDING** 891 E. Delos Santos Ave., Quezon City, Philippines 1103
Tel. Nos.: (632) 8634-0405; (632) 8924-2225 Fax: (632) 8924-2094 Email: molave@acerogroup.org

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**Technical Specifications:**

| | |
|---|---|
| **Capacity (Metric)** | 65 L (2.3 cu.ft.) |
| **Type** | Gravity Convection |
| **Controller Type** | Microprocessor control with vacuum fluorescent display |
| **Temperature (Metric)** | 50°C to 250°C |
| **Temperature Stability** | ±0.4°C |
| **Timer** | ON / OFF |
| **Dimensions (D x W x H) Interior** | 16.3 x 12.9 x 18.9 in. (41.5 x 32.8 x 48 cm) |
| **Dimensions (L x W x H)** | 22.2 x 20.9 x 28.3 in. (56.5 x 53 x 72 cm) |
| **Electrical Requirements** | 230 V 50/60 Hz, 1800W, 7.9 A |
| **No. of Shelves** | 2 supplied/13 max. |

**…….** **P 136,715.00/unit - F**

**ACERO BUILDING** 891 E. Delos Santos Ave., Quezon City, Philippines 1103
Tel. Nos.: (632) 8634-0405; (632) 8924-2225 Fax: (632) 8924-2094 Email: molave@acerogroup.org

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TRADING INC.

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TERMS OF PAYMENT: Immediate.
DELIVERY PERIOD : Within 60 -90 working days after receipt of your P.O.
VALIDITY : Within one hundred twenty (120) days after receipt of our offer thereafter subject to final confirmation.
WARRANTY : One (1) year from date of acceptance.
REMARKS : Standard VAT included
In case of cancellation of your order to us, kindly note the corresponding charges, as follows:
Within 15 days from issuance of P.O. – NO CHARGE
Within 30 days from issuance of P.O. – 75 %
Within 45 days from issuance of P.O. – 100 %

Thank you and we look forward to receiving your most valued order soon.

Yours very truly,
**MOLAVE TRADING, INC.**

**LOIDA B. VILLAMENA**
Department Manager
Life Science Group

MOLAVE
TRADING INC.

ISO 9001:2015

January 5, 2022

**PARTIDO STATE UNIVERSITY**
Camarines Sur

Attention: **ENGR. KIER P. GASGA**
Research and Development Coordinator

**Re: ESTABLISHMENT OF BIOASSY AND POTENTIAL DRUG SCREENING RESEARCH LABORATORY**

Dear Sir,

As the authorized Philippine distributor of Thermo Fisher Scientific, we are pleased to offer the ff:-

**Item No. 6**

**One (1) unit THERMO SCIENTIFIC MEGAFUGE 8 VENTILATED CENTRIFUGE CAT 75007211**

Technical Specifications:

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|---|---|
| Dimensions (HWD) | 12.2 x 14.6 x 18.9 in. (31 x 37 x 48 cm) |
| Drive System | Direct Brushless Induction |
| Profile (Acceleration/Braking) | 2 (standard and soft) |
| Program Storage | 4 protocols via direct access key, with naming option and password protection available |
| Run Time | 99 hr. 59 min., continuous |
| Capacity | 4 x 145 mL (TX-150 Swinging Bucket Rotor),<br>6 x 50 mL (HIGHConic III Fixed Angle Rotor<br>24 x 2ml ( Microclick Rotor) |
| Max. Speed | 4,500 rpm (TX-150 rotor),<br>16,000 rpm (MicroClick 24 x 2 rotor)<br>8,700 rpm ( High Conic III Rotor) |
| Max. RCF | 3,260 x g (TX-150 rotor),<br>24,328 x g (MicroClick 24 x 2 rotor)<br>10,155 xg ( High Conoc III Rotor) |
| Noise Level | <58 dB(A) (with TX-150 rotor); <61 dB(A) (with MicroClick 24 x 2 rotor) |
| Safety Features | Auto-Lock rotor exchange system, SMARTSpin imbalance detection system, certified ClickSeal biocontainment lids |
| Certifications/Compliance | IEC 61010-1, IEC 61010-2-020, IEC 61010-2-101, EN 61326-1; UL Listed / CE marked /IVD compliant / Biocontainment certified by the Public Health England, Porton Down, UK |
| Electrical Requirements | 230 V 50/60 Hz, 310W |
| Controller Type | Microprocessor control |

Accessories included:

1 x cat 75005760 Cell Culture Rotor Package includes:

1 x cat 75005701 TX-150 Swing Out Rotor

Max Rotor Speed: 4,500rpm Max Rotor RCF: 3,260 xg

Max Rotor Capacity: 4 x 145 ml

1 x cat 75005702 Round Buckets (set/4)

1 x cat 75005736 Adapters for 50 ml tubes (set/4 )

……..**P 366,775.00/unit**

Optional Accessories:

1 x cat 75005709 Hi Conic III Fixed Angle Rotor

Max Rotor Speed: 8,700 rpm Max Rotor RCF: 10,155 x g

Max Rotor Capacity: 6 x 50ml

……..**P 166,451.00/unit**

1 x cat 75005715 MicroClick Rotor 24 x 2ml

Max Rotor Speed: 16,000 rpm Max Rotor RCF: 24,328 x g

Max Rotor Capacity: 24 x 2ml

……..**P 93,998.00/unit**

**Item no. 7**

**One (1) pack THERMO SCIENTIFIC MULTIFUGE X4R Pro REFRIGERATED CENTRIFUGE cat 75016066**

**Technical Specifications:**

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|---|---|
| Description | Thermo Scientific Multifuge X4R Pro 220V-240V 50Hz/230V 60Hz, TX-1000 Cell Culture Package |
| Capacity | 96 x 15 mL Conical Tubes, 40 x 50 mL Conical Tubes, 6 Microplates, 4 x 1000 mL BioBottles (with TX-1000 rotors) |
| Max. RCF | 3,374 x g (TX-1000 Swing Out Rotor) |
| Max. Speed | 3,800 rpm (TX-1000 Swing Out Rotor) |
| Refrigerated | Yes, CFC-Free, Pre-Temp Available |
| Display | Full Color Touch Screen Interface |
| Timer | Option to count up or down |
| Certifications/Compliance | EN/UL/IEC 61010-1EN/UL/IEC 61010-2EN/UL/IEC 61326-1EN/UL/IEC 61326-2FCC part 15 |
| Includes | Printed English user manual, USB stick with 28 languages of user manual, corrosion resistance oil (70009824), power cord for the according country supply voltage, declaration of conformity |
| Electrical Requirements | 220/240 V 50 Hz, 230 V 60 Hz |

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|---|---|
| Controller Type | Microprocessor |
| Profile (Acceleration/Braking) | Choices of 9 Accel/10 Decel Rates |
| Drive System | Direct, Brushless Induction Low Profile |
| Program Storage | Up to 100 Programs, including multi-step programming via Touch Screen; Including programmabile Pulse options |
| Run Time | Up to 99 hr. 59 min. 59 sec. |
| Standards | ISO 14971, ISO 13485, ISO 9001 |
| Temperature Range | -10°C to +40°C, ±2°C |

**Product Overview:**

*Simple Operation and Design*

Full color intuitive touch screen for fast programming, centrifuge health monitoring and up to 100 program storage
Modern industrial design supports continuous improvement in rotor loading ergonomics, user safety and current needed certifications
Detailed operation information, including display of set and actual parameters, maximizes run performance and temperature accuracy for improved sample protection
Easily adjust speed, time and temperature, even during the run, with the convenient keyboard for direct input
Innovative lid-locking technology enables one finger downward motion for easy closing and locking of centrifuge lid, even on tall benches
Double temperature management system displays the air temperature in the chamber as well as the sample temperature, ensuring accurate control of the sample temperature during the run

*Thermo Scientific™ Auto-Lock™ III Rotor System*

Secure locking system allows easy push-button installation and exchange of rotors
Saves time in switch between applications
Easy access to the chamber enables quick cleaning, a healthier working environment and longer unit life

*Thermo Scientific™ ClickSeal™ Bucket Sealing Syste*m

Biocontainment system seals with a snap; eliminates screw caps and complicated clips
Glove-friendly, one-handed open/close capability
Certified by CAMR™ in Porton Down, UK

*Thermo Scientific™ SMARTSpin™ Technology*

Advanced rotor management system maximizes acceleration, braking and residual load imbalance for each rotor and bucket option
Optimizes safety and improves separations

*Space-Saving Design*

Compact size conserves workspace on benchtop or floor
Ergonomic height and curved design ensure comfortable sample handling and cleaning of unit
Energy Efficient
Design saves up to 40% of energy on industry standard protocols such as blood separations or conical-tube processing
Programmable centrifuge compressor start and stop can save both time and energy

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Accessories included in the package:

1 x TX-1000 Swing Out Rotor
1 x Buckets (set/4)
1 x Clicks Seal Biocontainment Lids (set/4)
1 x Adapters for 50ml tubes ( set /4 , 40 x 50ml tubes)
1 x Adapters for 15 ml tubes (set /4, 96 x 15 ml tubes)

……P 905,000.00/package

A. Optional Accessories for use with TX-1000 Swing Out Rotor

1 x Adapters for 196 x 5/7 ml tubes cat 75003671
1 x Adapters for 148 x 9/10ml tubes cat 75003697
1 x Adapters for 1000ml bottles(pk/4) cat 75007301

……P 181,817.00/package

B. Optional Rotors for the Multifuge X4R Pro Refrigerated Centrifuge

1 x Microliter Rotor 24 x 1.5/2 ml Fixed Angle Rotor cat 75003652
Max Rotor Speed: 15,200 rpm Max RCF: 25,830 x g
……P 88,551.00/unit

1 x Bioshield 1000A Rotor cat 75003182
Max Rotor Speed: 6,000 rpm Max Rotor RCF: 7,164 xg
Max Rotor Capacity: 4 x 250ml
1 x Adapters for 4 x 240ml Flat Bottom Bottle cat 75003737
……P 365,455.00/unit

| | |
|---|---|
| TERMS OF PAYMENT: | Immediate upon receipt of goods. |
| DELIVERY: | Within 90-120 days after receipt of your approved P.O. |
| WARRANTY: | One (1) year from date of acceptance. |
| INSTALLATIONS/TRAINING: | To be done on site by our factory trained service engineers. |
| REMARKS: | Standard VAT included. |

Thank you and we look forward to receiving your most valued order soon.

Yours very truly,
**MOLAVE TRADING, INC.**

**LOIDA B. VILLAMENA**
Department Sales Manager
Life Science Group